Application Deployment

(Deploy the given React application to a production ready state)

Application:

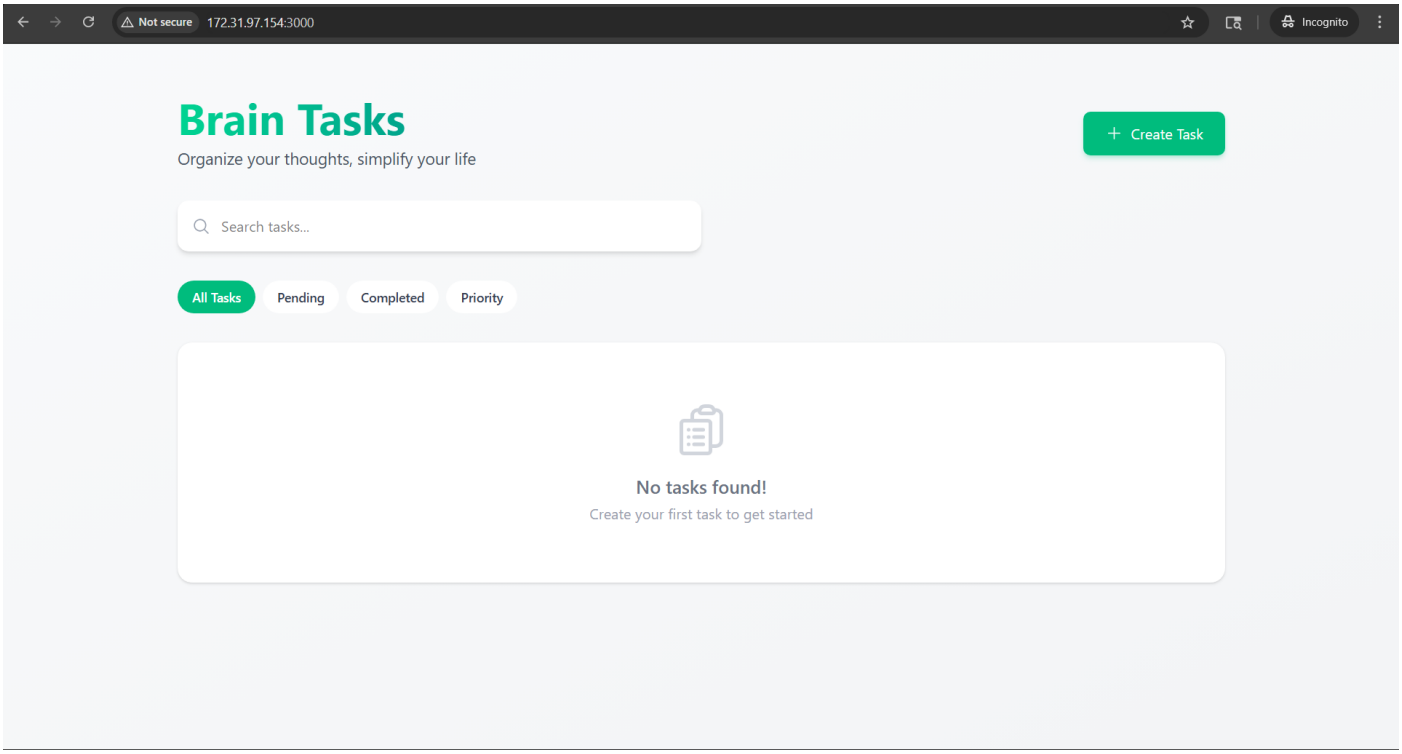
Clone the below mentioned repository and deploy the application (Run application in port 3000).

Repo URL: https://github.com/Vennilavanguvi/Brain-Tasks-App.git

Docker:

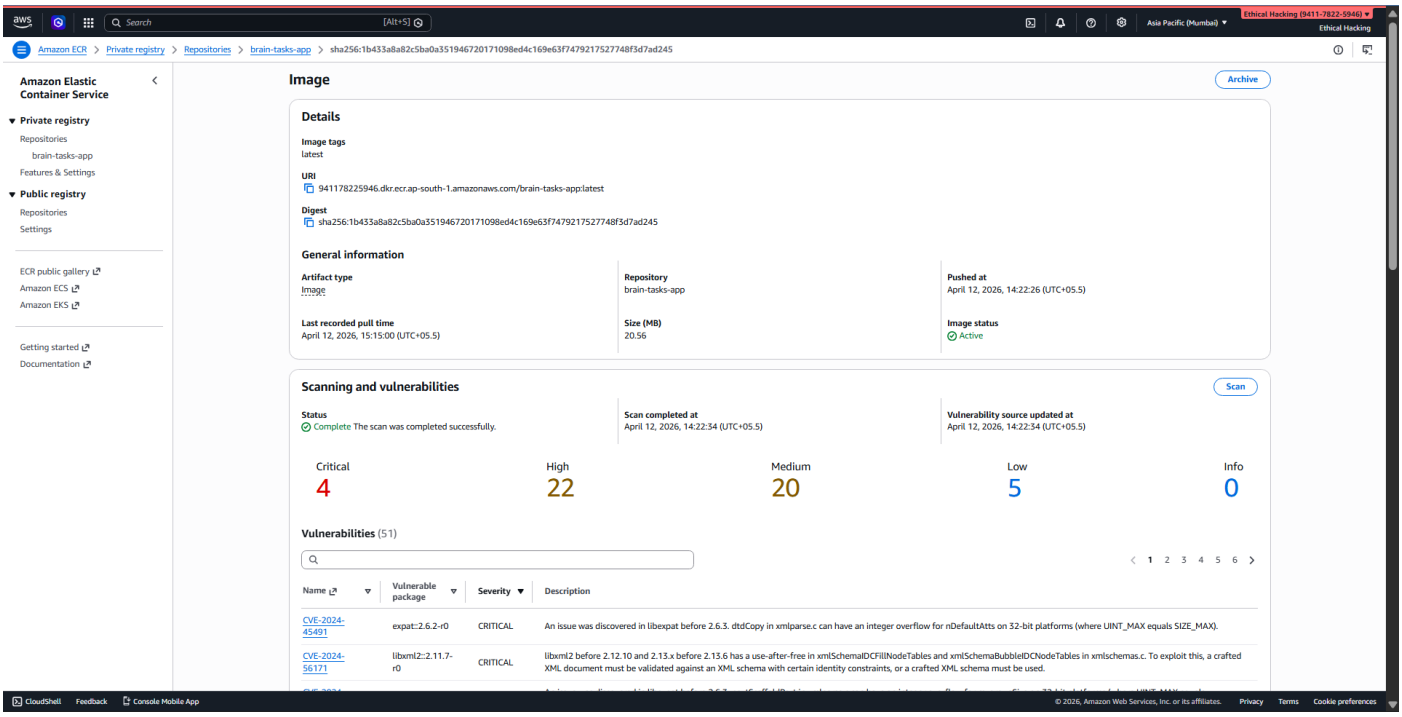
- Dockerize the application by creating Dockerfile
- Build an application and check output using docker image.

```
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ docker images
REPOSITORY                                                  TAG           IMAGE ID       CREATED         SIZE
941178225946.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app   latest        2e4f82804a9e   3 hours ago     48.6MB
brain-tasks-app                                             latest        2e4f82804a9e   3 hours ago     48.6MB
ghostatdocker/brain-tasks-app                               latest        778a8b93d719   2 days ago      62.5MB
hello-world                                                 latest        1b44b5a3e06a   8 months ago    10.1kB
nginx                                                       1.25-alpine   501d84f5d064   23 months ago   48.3MB
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ docker run -d -p 3000:3000 --name Brain_Task_Docker 941178225946.dkr.ecr.ap-south-1.amazonaws.com/b
rain-tasks-app
1c10505fa7e36d2b03367f6f0b966cfa34da3b34ff9ea07caeeb0c2c2045af92
```



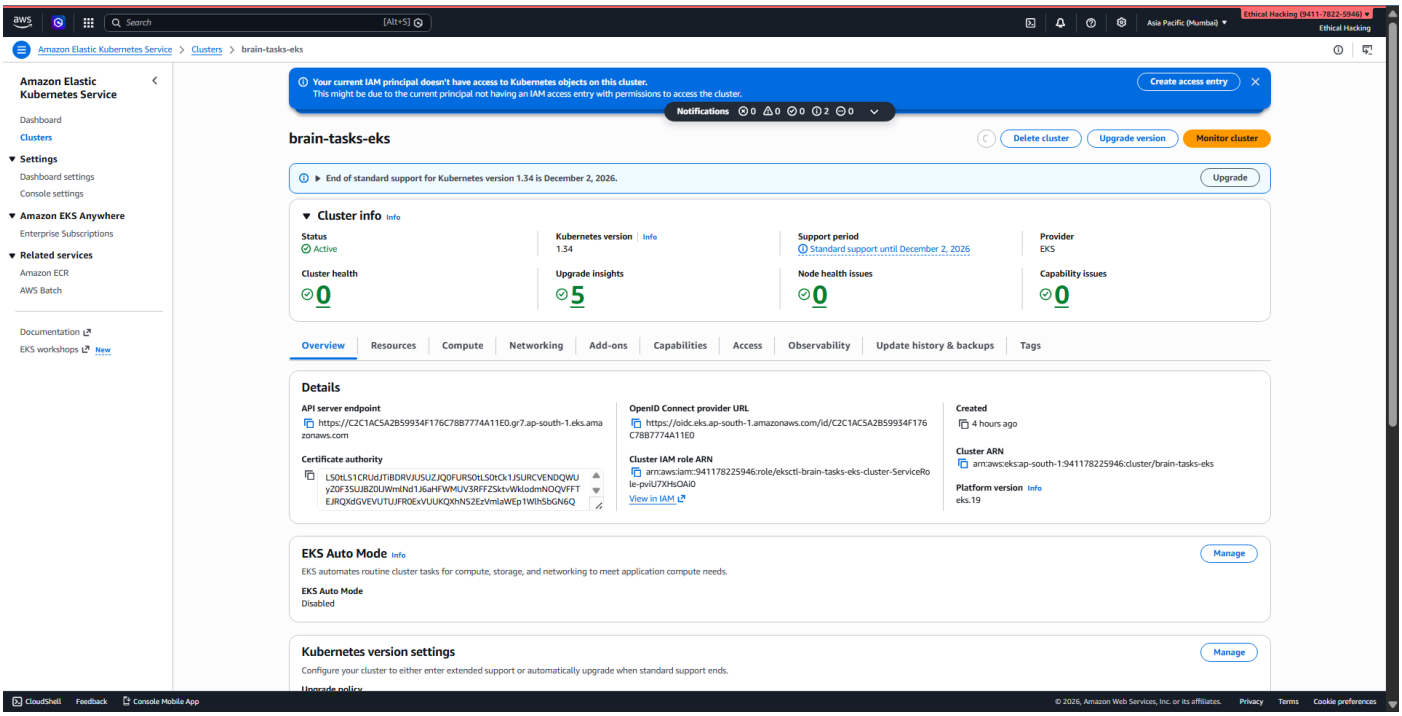
Registry:

- Create an AWS ECR or Dockerhub repository for storing docker images.



Kubernetes:

- Setup Kubernetes in AWS EKS and Confirm EKS cluster is running.



- Write deployment and service YAML files.

GHOST@GHOST: ~/Final_Proj Ubuntu

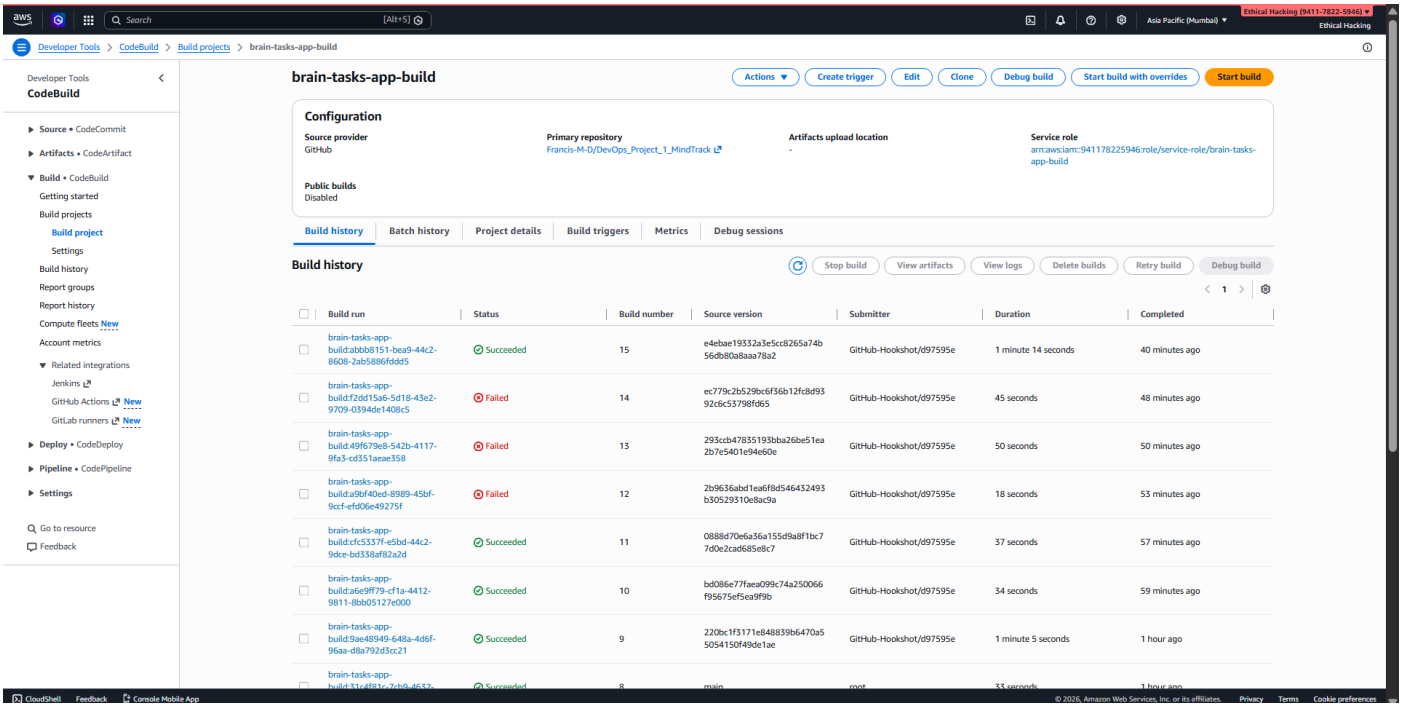
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-app
  namespace: production
  labels:
    app: brain-tasks-app
    version: "1.0.0"
    environment: production
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks-app
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0          # Zero-downtime rolling update
  template:
    metadata:
      labels:
        app: brain-tasks-app
        version: "1.0.0"
      annotations:
        prometheus.io/scrape: "false"
    spec:
      # Spread pods across AZs for HA
      topologySpreadConstraints:
        - maxSkew: 1
          topologyKey: topology.kubernetes.io/zone
          whenUnsatisfiable: DoNotSchedule
          labelSelector:
            matchLabels:
              app: brain-tasks-app

      containers:
        - name: brain-tasks-app
          # Updated by CodeBuild/CodePipeline at deploy time
          image: 941178225946.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
          imagePullPolicy: Always
          ports:
            - containerPort: 3000
"deployment.yaml" 112L, 2694B
```

GHOST@GHOST: ~/Final_Proj Ubuntu

```yaml
          env:
            - name: NGINX_PORT
              value: "3000"
            - name: APP_ENV
              value: "production"

          # Read-only root filesystem for security
          securityContext:
            readOnlyRootFilesystem: false
            runAsNonRoot: false
            allowPrivilegeEscalation: false

      restartPolicy: Always
      terminationGracePeriodSeconds: 30

---
# Horizontal Pod Autoscaler
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: brain-tasks-app-hpa
  namespace: production
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: brain-tasks-app
  minReplicas: 2
  maxReplicas: 6
  metrics:
    - type: Resource
      resource:
        name: cpu
        target:
          type: Utilization
          averageUtilization: 70
    - type: Resource
      resource:
        name: memory
        target:
          type: Utilization
          averageUtilization: 80
"deployment.yaml" 112L, 2694B
```

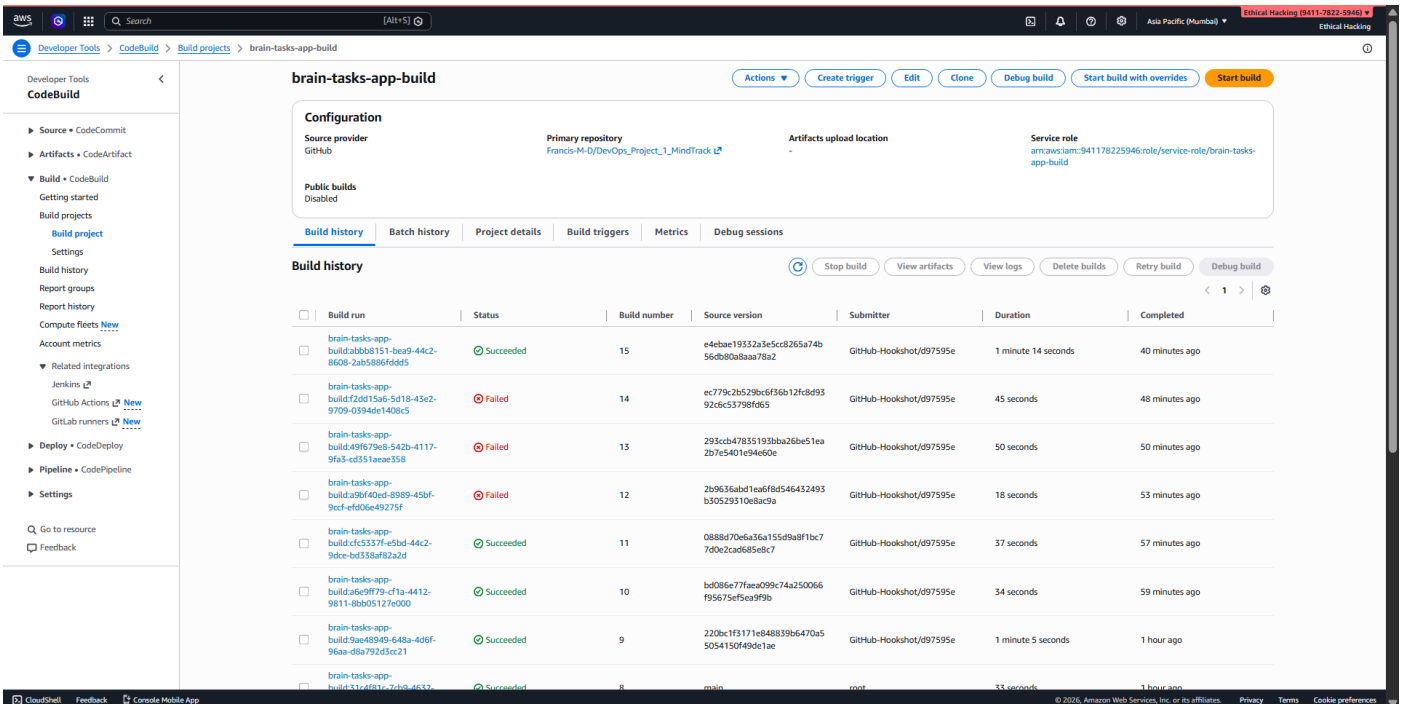
```yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-app-svc
  namespace: production
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks-app
  ports:
    - port: 80
      targetPort: 3000
~
"service.yaml" 12L, 202B
```

- Deploy using kubectl via CodeBuild or Codepipeline.

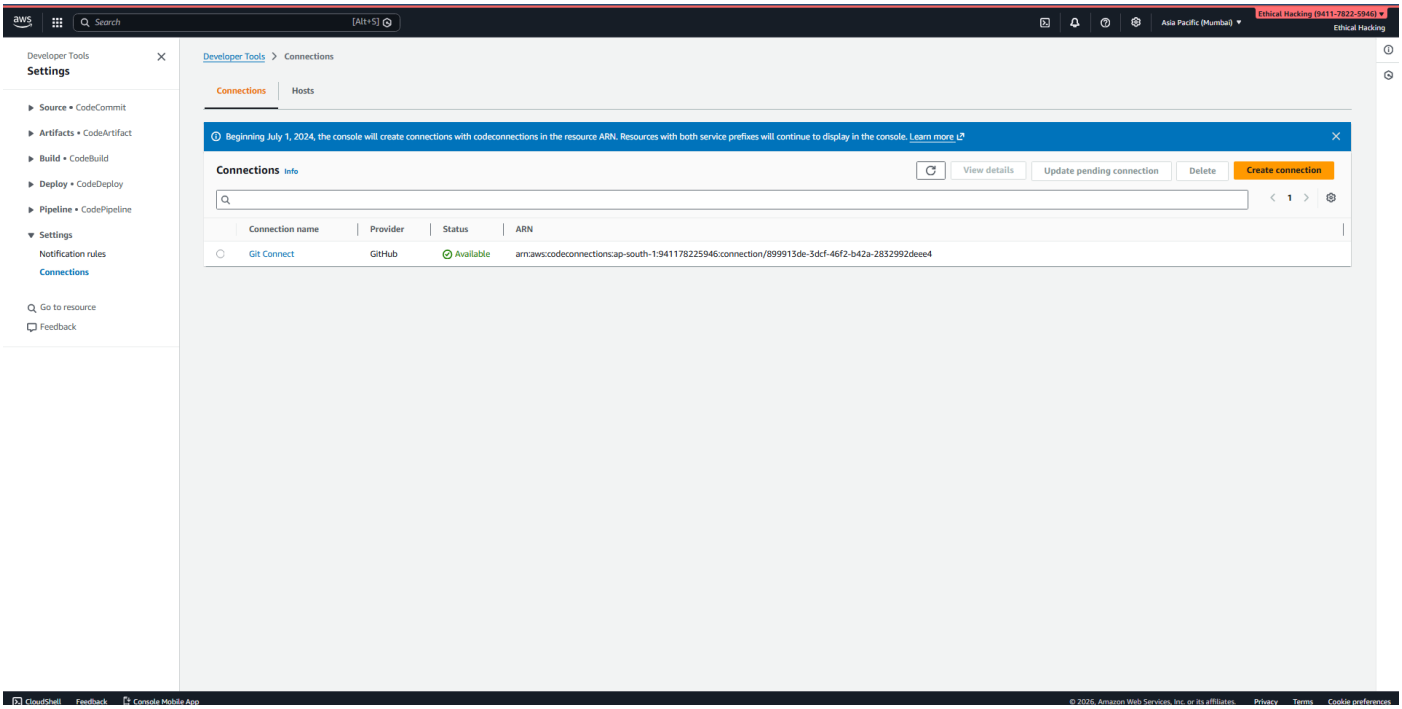


CodeBuild:

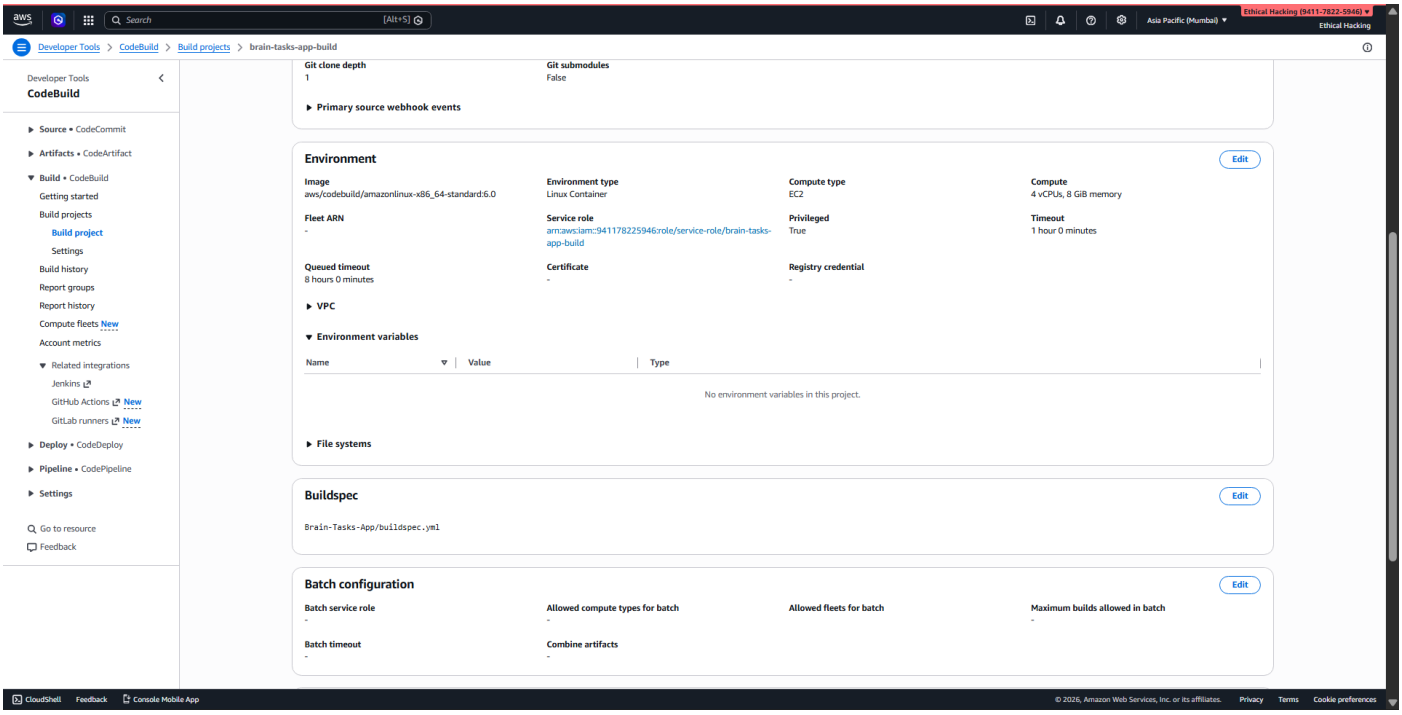
- Create a CodeBuild project:



- Source: Connect to your repository



- Environment: Use managed image (Amazon Linux, Ubuntu)

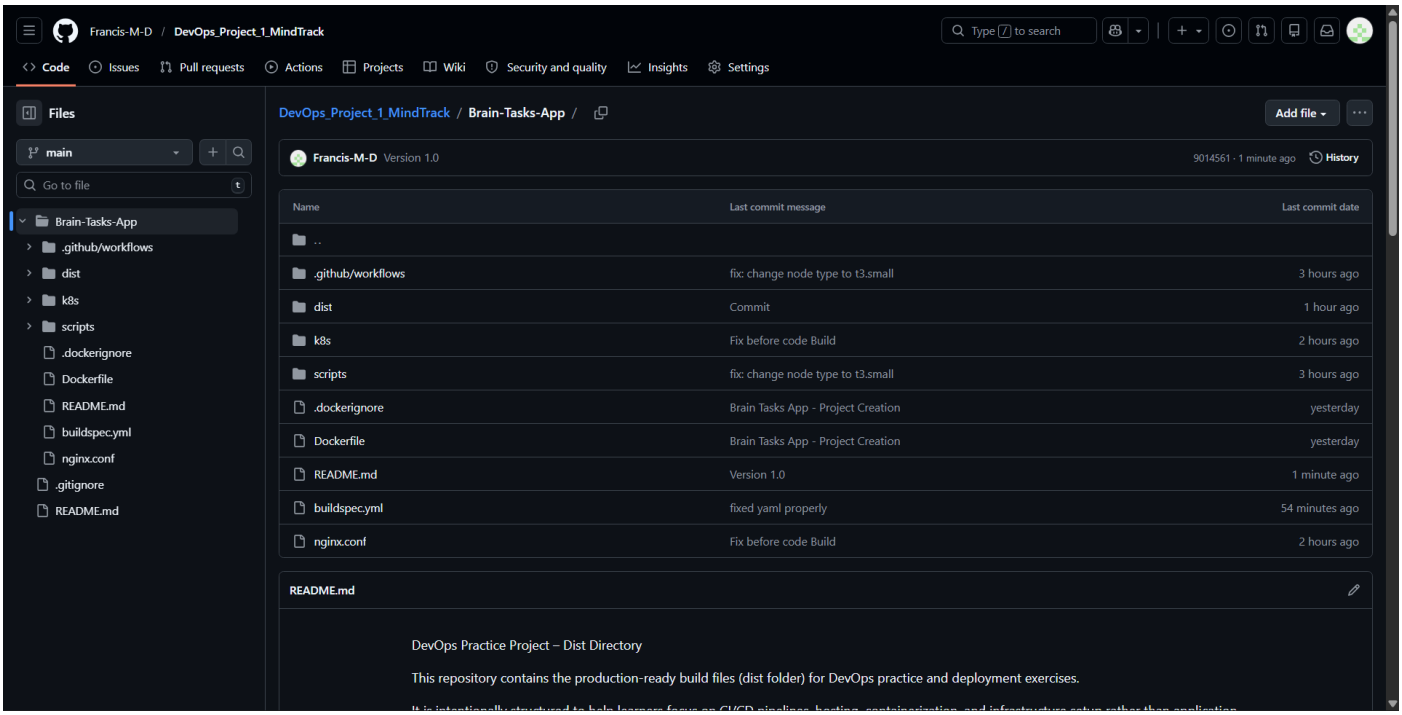


- Write and define commands in buildspec.yml.

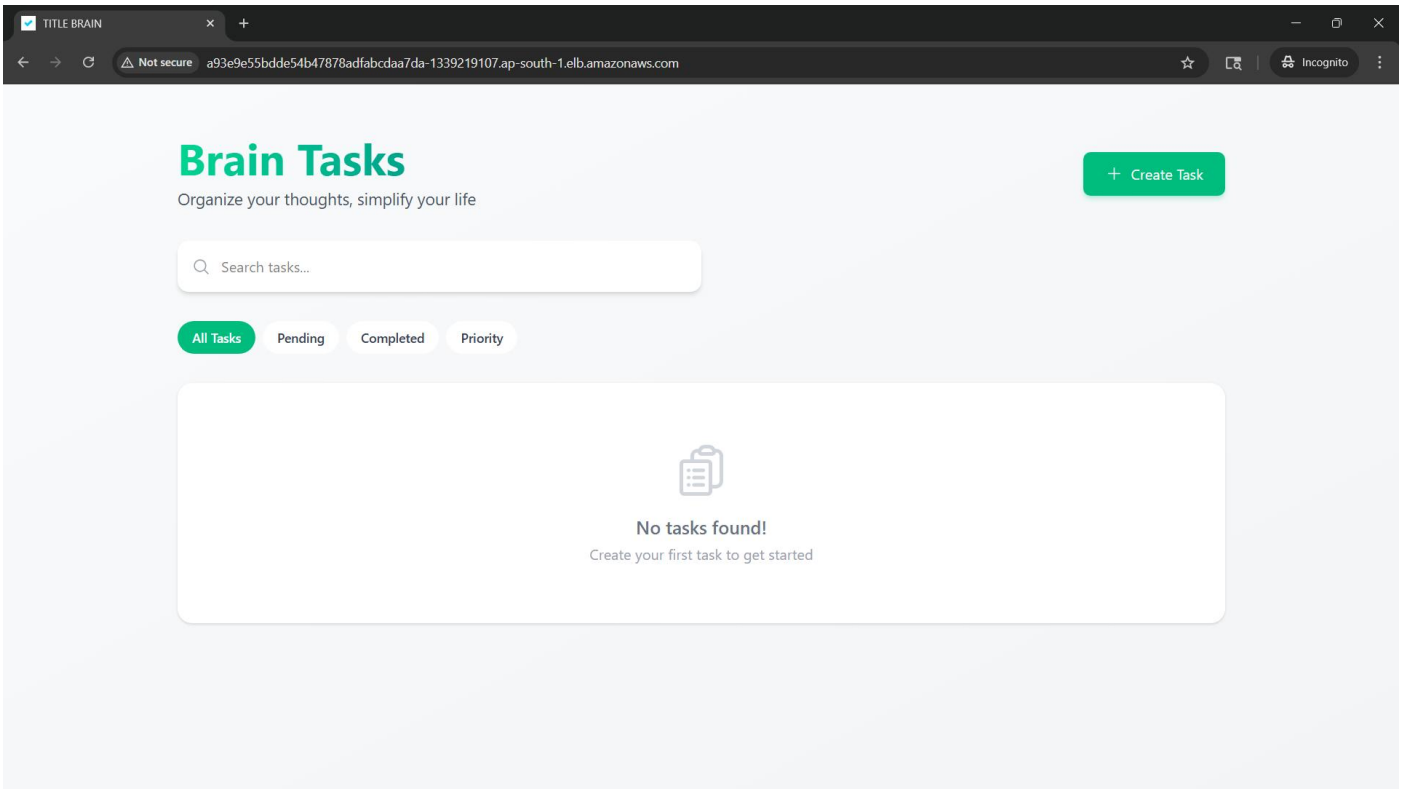
```yaml
version: 0.2

env:
  variables:
    AWS_REGION: "ap-south-1"
    ACCOUNT_ID: "941178225946"
    IMAGE_REPO_NAME: "brain-tasks-app"
    CLUSTER_NAME: "brain-tasks-eks"

phases:
  pre_build:
    commands:
      - echo "Logging in to Amazon ECR"
      - aws ecr get-login-password --region $AWS_REGION | docker login --username AWS --password-stdin $ACCOUNT_ID.dkr.ecr.$AWS_REGION.amazonaws.com

  build:
    commands:
      - echo "Build started"
      - IMAGE_TAG=$(date +%s)
      - echo "Building Docker image"
      - docker build -t $IMAGE_REPO_NAME:$IMAGE_TAG ./Brain-Tasks-App
      - docker tag $IMAGE_REPO_NAME:$IMAGE_TAG $ACCOUNT_ID.dkr.ecr.$AWS_REGION.amazonaws.com/$IMAGE_REPO_NAME:$IMAGE_TAG

  post_build:
    commands:
      - echo "Pushing image"
      - docker push $ACCOUNT_ID.dkr.ecr.$AWS_REGION.amazonaws.com/$IMAGE_REPO_NAME:$IMAGE_TAG
      - echo "Updating kubeconfig"
      - aws eks update-kubeconfig --region $AWS_REGION --name $CLUSTER_NAME
      - echo "Deploying to Kubernetes"
      - kubectl set image deployment/brain-tasks-app brain-tasks-app=$ACCOUNT_ID.dkr.ecr.$AWS_REGION.amazonaws.com/$IMAGE_REPO_NAME:$IMAGE_TAG -n production
      - kubectl rollout status deployment/brain-tasks-app -n production
~
~
~
~
~
~
~
~
~
~
~
"buildspec.yml" 32L, 1234B                                                    32,71         All
```

Version Control:

- Push the codebase to a Git provider (GitHub).



- Use CLI commands to push code.
    - Current Output, I am going to change the content of the website [The current Website Title is “TITLE BRAIN”].




```
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ cat Brain-Tasks-App/dist/index.html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="https://is1-ssl.mzstatic.com/image/thumb/Purple221/v4/1b/a6/70/1ba67006-9749-9567-151b-731d5cba4f93/AppIcon-
0-0-1x_U007epad-0-1-0-85-220.jpeg/512x512bb.jpg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>TITLE BRAIN</title>
    <script type="module" crossorigin src="/assets/index-BHGiHu50.js"></script>
    <link rel="stylesheet" crossorigin href="/assets/index-DPTLVrPB.css">
  </head>
  <body>
    <div id="root"></div>
  </body>
</html>
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ sed -i 's/TITLE BRAIN/BRAIN TASK/g' Brain-Tasks-App/dist/index.html
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ cat Brain-Tasks-App/dist/index.html | grep -i "BRAIN TASK"
    <title>BRAIN TASK</title>
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Brain-Tasks-App/dist/index.html

no changes added to commit (use "git add" and/or "git commit -a")
```

```
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ git add .
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ git commit -m "Changing Website Title"
[main c14a8b4] Changing Website Title
 1 file changed, 1 insertion(+), 1 deletion(-)
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 551 bytes | 275.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Francis-M-D/DevOps_Project_1_MindTrack.git
   9014561..c14a8b4  main -> main
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$
```

Developer Tools > CodeBuild > Build projects > brain-tasks-app-build > brain-tasks-app-build:9ffcb3a7-7536-4173-be6b-7b5da7d7cc7e

brain-tasks-app-build:9ffcb3a7-7536-4173-be6b-7b5da7d7cc7e

Stop build | Debug build | Retry build

Build status

Status	Initiator	Build ARN	Resolved source version
In progress	GitHub-Hookshot/d97595e	arn:aws:codebuild:ap-south-1:941178225946:build/brain-tasks-app-build:9ffcb3a7-7536-4173-be6b-7b5da7d7cc7e	c14a8b410ada412a0aac785b091e4c4a35d8f4dc
Start time	**End time**	**Build number**	
Apr 12, 2026 4:22 PM (UTC+5:30)	-	17	

Build logs | Phase details | Reports | Environment variables | Build details | Resource utilization

Showing the last 27 lines of the build log. View entire log — Tail logs

```
[Container] 2026/04/12 10:52:40.781575 Running on CodeBuild On-demand
[Container] 2026/04/12 10:52:40.781589 Waiting for agent ping
[Container] 2026/04/12 10:52:42.287869 Waiting for DOWNLOAD_SOURCE
[Container] 2026/04/12 10:52:48.126645 Phase is DOWNLOAD_SOURCE
[Container] 2026/04/12 10:52:48.128380 CODEBUILD_SRC_DIR=/codebuild/output/src56792885/src/github.com/Francis-M-D/DevOps_Project_1_MindTrack
[Container] 2026/04/12 10:52:48.128946 YAML location is /codebuild/output/src56792885/src/github.com/Francis-M-D/DevOps_Project_1_MindTrack/Brain-Tasks-App/buildspec.yml
[Container] 2026/04/12 10:52:48.131064 Setting HTTP client timeout to higher timeout for Github and GitHub Enterprise sources
[Container] 2026/04/12 10:52:48.131139 Processing environment variables
[Container] 2026/04/12 10:52:49.466068 No runtime version selected in buildspec.
[Container] 2026/04/12 10:52:49.531272 Moving to directory /codebuild/output/src56792885/src/github.com/Francis-M-D/DevOps_Project_1_MindTrack
[Container] 2026/04/12 10:52:49.531295 Cache is not defined in the buildspec
[Container] 2026/04/12 10:52:50.174334 Skip cache due to: no paths specified to be cached
[Container] 2026/04/12 10:52:50.174640 Registering with agent
[Container] 2026/04/12 10:52:50.794943 Phases found in YAML: 3
[Container] 2026/04/12 10:52:50.794966  PRE_BUILD: 2 commands
[Container] 2026/04/12 10:52:50.794971  BUILD: 5 commands
[Container] 2026/04/12 10:52:50.794975  POST_BUILD: 7 commands
[Container] 2026/04/12 10:52:50.795289 Phase complete: DOWNLOAD_SOURCE State: SUCCEEDED
[Container] 2026/04/12 10:52:50.795302 Phase context status code:  Message:
[Container] 2026/04/12 10:52:52.401117 Entering phase INSTALL
[Container] 2026/04/12 10:52:53.051103 Phase complete: INSTALL State: SUCCEEDED
[Container] 2026/04/12 10:52:53.051124 Phase context status code:  Message:
[Container] 2026/04/12 10:52:53.804111 Entering phase PRE_BUILD
[Container] 2026/04/12 10:52:53.763991 Running command echo "Logging in to Amazon ECR"
Logging in to Amazon ECR

[Container] 2026/04/12 10:52:53.772418 Running command aws ecr get-login-password --region $AWS_REGION | docker login --username AWS --password-stdin $ACCOUNT_ID.dkr.ecr.$AWS_REGION.amazonaws.com
```

Developer Tools > CodeBuild > Build projects > brain-tasks-app-build > brain-tasks-app-build:9ffcb3a7-7536-4173-be6b-7b5da7d7cc7e

brain-tasks-app-build:9ffcb3a7-7536-4173-be6b-7b5da7d7cc7e

Stop build | Debug build | Retry build

Build status

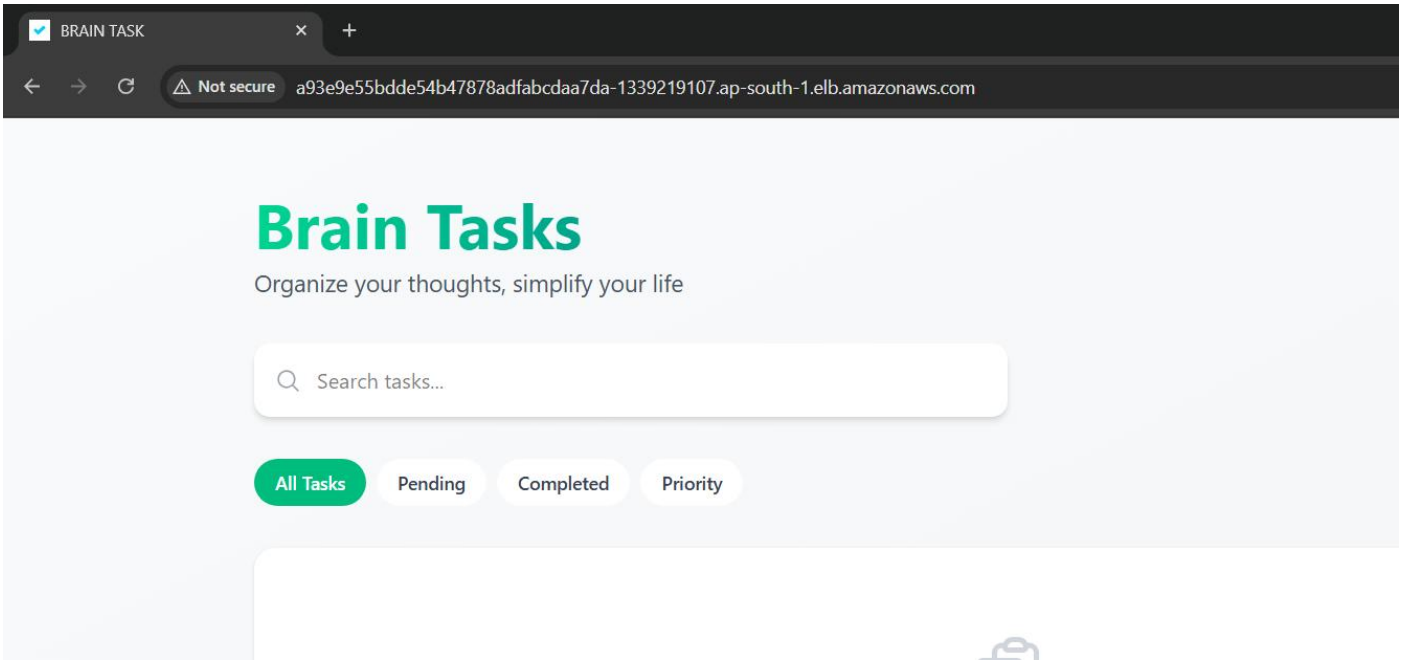
Status	Initiator	Build ARN	Resolved source version
Succeeded	GitHub-Hookshot/d97595e	arn:aws:codebuild:ap-south-1:941178225946:build/brain-tasks-app-build:9ffcb3a7-7536-4173-be6b-7b5da7d7cc7e	c14a8b410ada412a0aac785b091e4c4a35d8f4dc
Start time	**End time**	**Build number**	
Apr 12, 2026 4:22 PM (UTC+5:30)	Apr 12, 2026 4:23 PM (UTC+5:30)	17	

Build logs | Phase details | Reports | Environment variables | Build details | Resource utilization

Showing the last 187 lines of the build log. View entire log — Tail logs

No previous logs

```
[Container] 2026/04/12 10:52:40.781575 Running on CodeBuild On-demand
[Container] 2026/04/12 10:52:40.781589 Waiting for agent ping
[Container] 2026/04/12 10:52:42.287869 Waiting for DOWNLOAD_SOURCE
[Container] 2026/04/12 10:52:48.126645 Phase is DOWNLOAD_SOURCE
[Container] 2026/04/12 10:52:48.128380 CODEBUILD_SRC_DIR=/codebuild/output/src56792885/src/github.com/Francis-M-D/DevOps_Project_1_MindTrack
[Container] 2026/04/12 10:52:48.128946 YAML location is /codebuild/output/src56792885/src/github.com/Francis-M-D/DevOps_Project_1_MindTrack/Brain-Tasks-App/buildspec.yml
[Container] 2026/04/12 10:52:48.131064 Setting HTTP client timeout to higher timeout for Github and GitHub Enterprise sources
[Container] 2026/04/12 10:52:48.131139 Processing environment variables
[Container] 2026/04/12 10:52:49.466068 No runtime version selected in buildspec.
[Container] 2026/04/12 10:52:49.531272 Moving to directory /codebuild/output/src56792885/src/github.com/Francis-M-D/DevOps_Project_1_MindTrack
[Container] 2026/04/12 10:52:49.531295 Cache is not defined in the buildspec
[Container] 2026/04/12 10:52:50.174334 Skip cache due to: no paths specified to be cached
[Container] 2026/04/12 10:52:50.174640 Registering with agent
[Container] 2026/04/12 10:52:50.794943 Phases found in YAML: 3
[Container] 2026/04/12 10:52:50.794966  PRE_BUILD: 2 commands
[Container] 2026/04/12 10:52:50.794971  BUILD: 5 commands
[Container] 2026/04/12 10:52:50.794975  POST_BUILD: 7 commands
[Container] 2026/04/12 10:52:50.795289 Phase complete: DOWNLOAD_SOURCE State: SUCCEEDED
[Container] 2026/04/12 10:52:50.795302 Phase context status code:  Message:
[Container] 2026/04/12 10:52:52.401117 Entering phase INSTALL
[Container] 2026/04/12 10:52:53.051103 Phase complete: INSTALL State: SUCCEEDED
[Container] 2026/04/12 10:52:53.051124 Phase context status code:  Message:
[Container] 2026/04/12 10:52:53.804111 Entering phase PRE_BUILD
[Container] 2026/04/12 10:52:53.763991 Running command echo "Logging in to Amazon ECR"
Logging in to Amazon ECR

[Container] 2026/04/12 10:52:53.772418 Running command aws ecr get-login-password --region $AWS_REGION | docker login --username AWS --password-stdin $ACCOUNT_ID.dkr.ecr.$AWS_REGION.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```



CodePipeline:

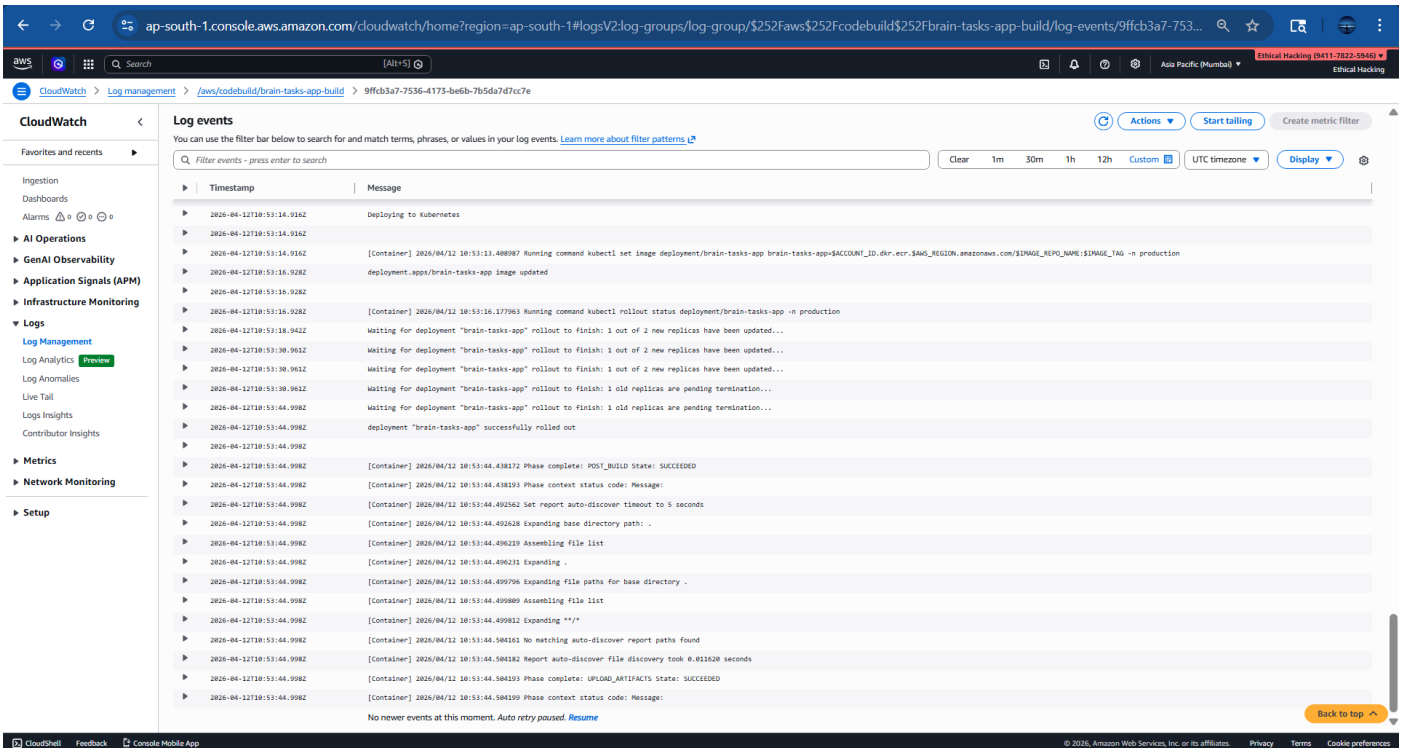
- Source: GitHub

```
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ git add .
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ git commit -m "Changing Website Title"
[main c14a8b4] Changing Website Title
 1 file changed, 1 insertion(+), 1 deletion(-)
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 551 bytes | 275.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Francis-M-D/DevOps_Project_1_MindTrack.git
   9014561..c14a8b4  main -> main
GHOST@GHOST:~/Final_Projects/DevOps_Project_1_MindTrack$ |
```

- Build: AWS CodeBuild project
- Deploy: Deploy to EKS via Codepipeline EKS deploy stage..

Monitoring:

- Use CloudWatch Logs to track build, deploy, and application logs.



Submission Guidelines:

- GitHub Link: Submit full code repository.

  https://github.com/Francis-M-D/DevOps_Project_1_MindTrack

- README File: Include setup instructions, pipeline explanation, and screenshots or create a document and attach all screenshots.

- Application deployed kubernetes Loadbalancer ARN.

  http://a93e9e55bdde54b47878adfabcdaa7da-1339219107.ap-south-1.elb.amazonaws.com/